



TSMC 0.18um USB 2.0 I/O Library

tpd973g_usb2

Application Note

v250b

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Taiwan Semiconductor Manufacturing Co., Ltd.

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REVISION HISTORY

Version	Date	Special Note
100	2003/2/22	1. First release
110	2003/11/12	1. Update layout guideline
120	2004/ 2/26	1. Add pad sequence 2. Add the BIST description 3. Add Verilog usage 4. Add preceding test in miscellaneous chapter 5. Correct some diagrams 6. Add the suspend mode section 7. Fix the floorplan typo
130	2005/11/22	1. Add the suspend mode between 250a and before.
250b	2005/12/9	1. Provide dedicated application note for tpd973g_usb2_250b library 2. Add Section 5.2 description 3. Remove redundant information for old library version

1 MACRO PLACEMENT

1.1 USB2.0 Transceiver Macrocell Floorplan

Fig-1 shows the floorplan of *tpd973g_usb2* USB2.0 transceiver macrocell (UTM). The red line encloses the macro phantom boundary. Note that the power cut cells on the end of I/O section is included in the macrocell, users need no additional layout work besides the I/O alignment and bonding pad insertion. Please DO NOT put any logo or polygon in the region below band-gap circuit to avoid short problem after the layout merge.

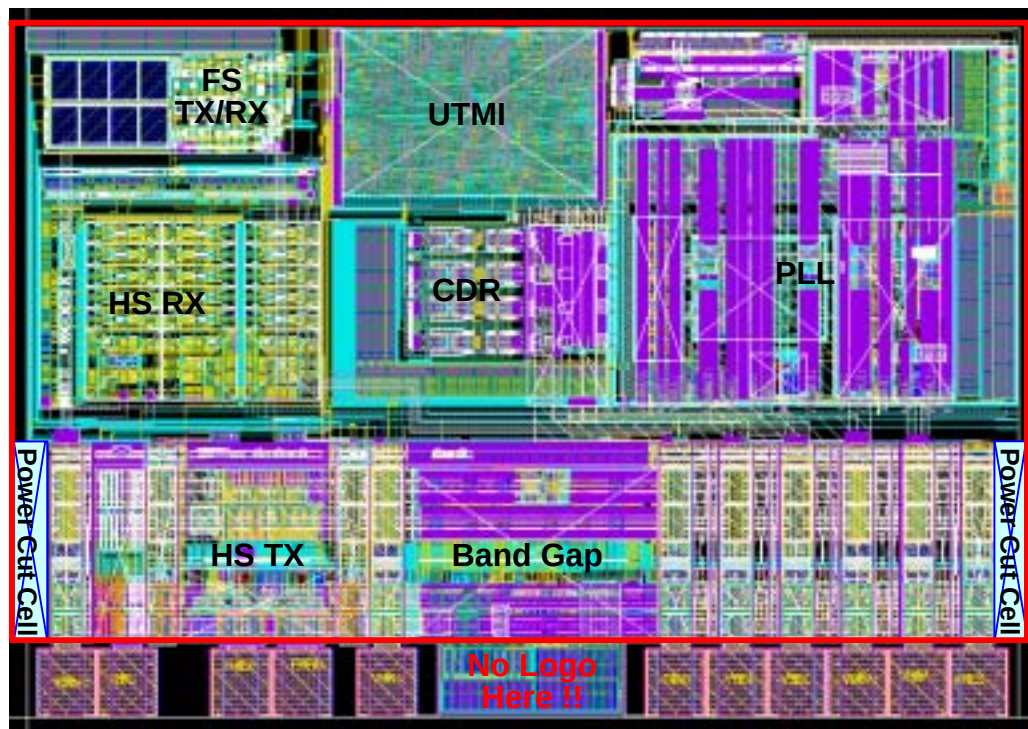


Fig-1 Floorplan of USB 2.0 transceiver macrocell (UTM)

1.2 Power/Ground Cells

Fig-2 shows the UTM placement example for a stand-alone UTMI PHY chip. User can easily align TSMC standard I/O cells with the macro power-cut cells and create the I/O ring. Besides the in-line bonding style shown in this example, users can adopt the staggered bonding style, which is also called EZBOND™.

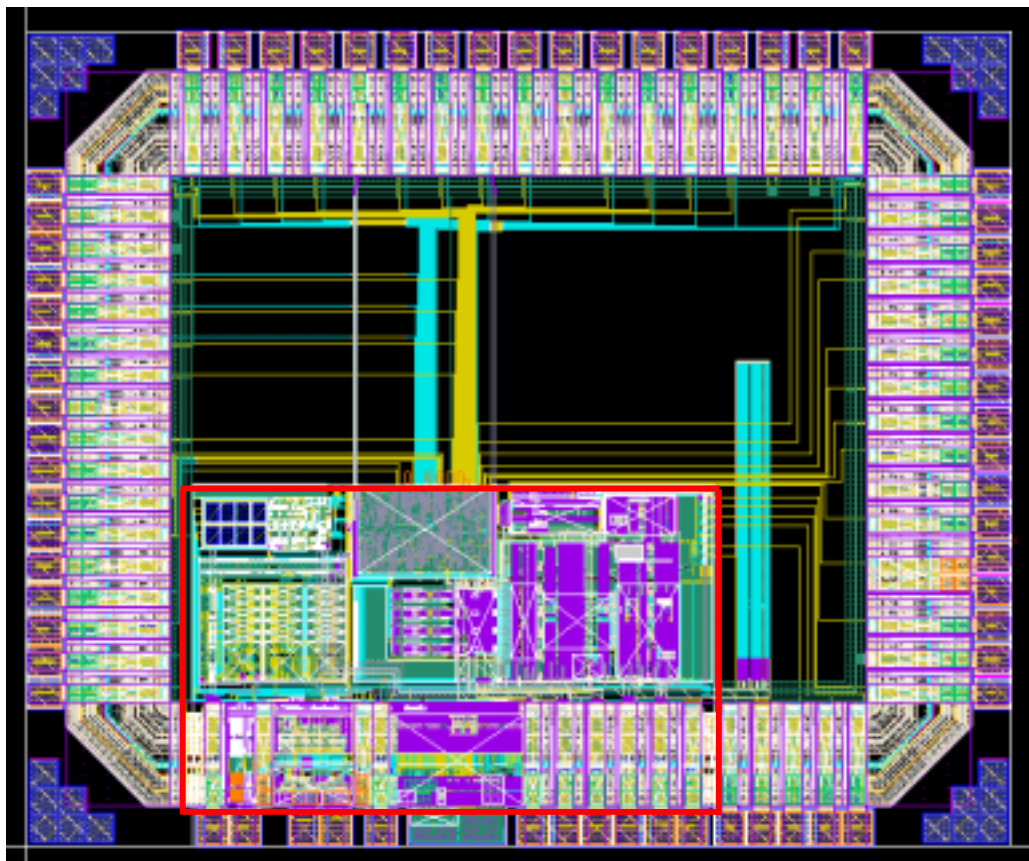


Fig-2 UTM placement example of a stand-alone chip

2 BI-DIRECTIONAL INTERFACE

2.1 Bi-directional 8-bit Interface

The UTM of *tpd973g_usb2* provides both 8-bit and 16-bit uni-directional bus interface. For different SIE interface, users may need bi-directional interface. Fig-3 shows the example to implement the 8-bit bi-directional data bus interface with TSMC standard I/O cells as a single chip. Users can use core cells to duplicate this structure. Please note that “**Databus16_8**” should be connected to “low” in this application.

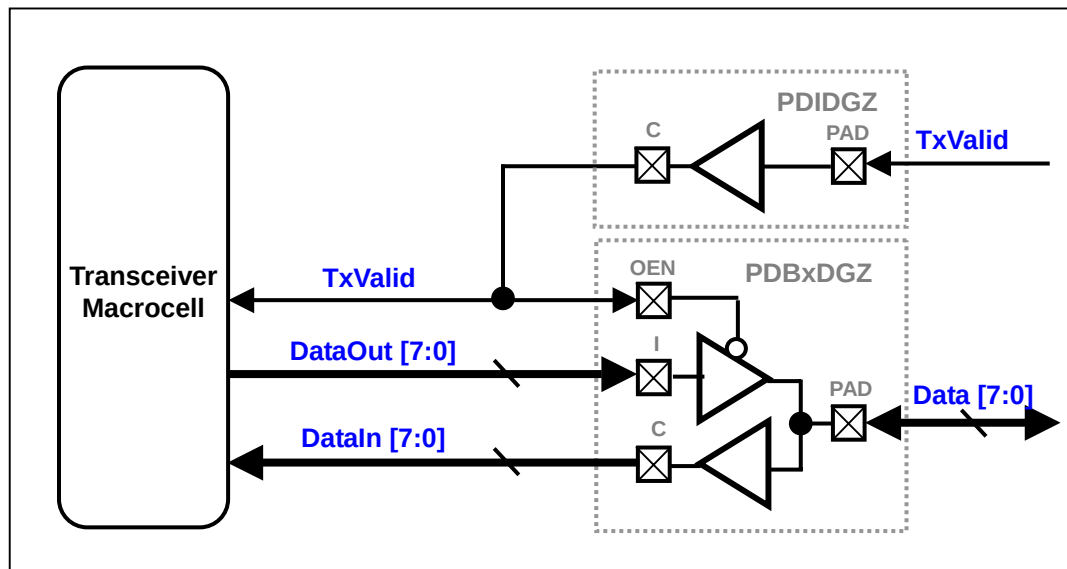


Fig-3 Implementation of bi-directional 8-bit bus interface

2.2 Bi-directional 16-bit Interface

Fig-4 shows the example to implement the 16-bit bi-directional data bus interface with TSMC standard I/O cells for a single-chip UTM. Note that “**DataBus16_8**” should be connected to “high” in this application. For users to implement both 8-bit uni-directional interface and 16-bit bi-directional interface in their single chip UTM, please refer to the configuration shown in Figure 19 of “USB2.0 Transceiver Macrocell Interface (UTMI) Specification Version 1.05”.

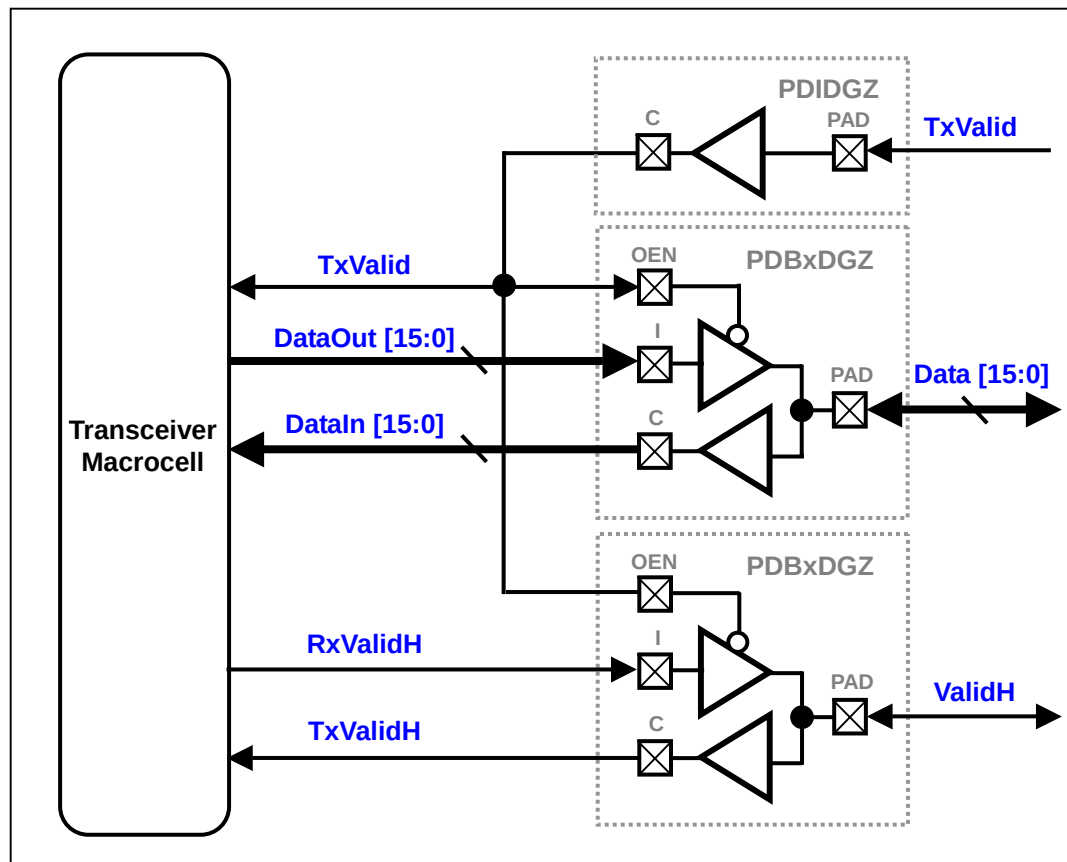


Fig-4 Implementation of bi-dir. 16-bit bus interface with TSMC standard

3 BIST APPLICATION

3.1 BIST Function Diagram

In USB2.0 high-speed transceiver mode, the operating frequency is up to 480MHz, combining both digital and analog part of circuit, that is hard to do mass production test in conventional digital test machine. Some users may choose high-speed tester, but those test usually need higher cost. TSMC provides the patented built-in-self-test (BIST) solution to solve this problem with small area and power overhead. This test method facilitates users to run real-time loop-back test in a low frequency (same to CLK) on a pure digital test machine. Only two extra pins are introduced: **BISTOn** to enable BIST and **BISTFail** to indicate chip failure.

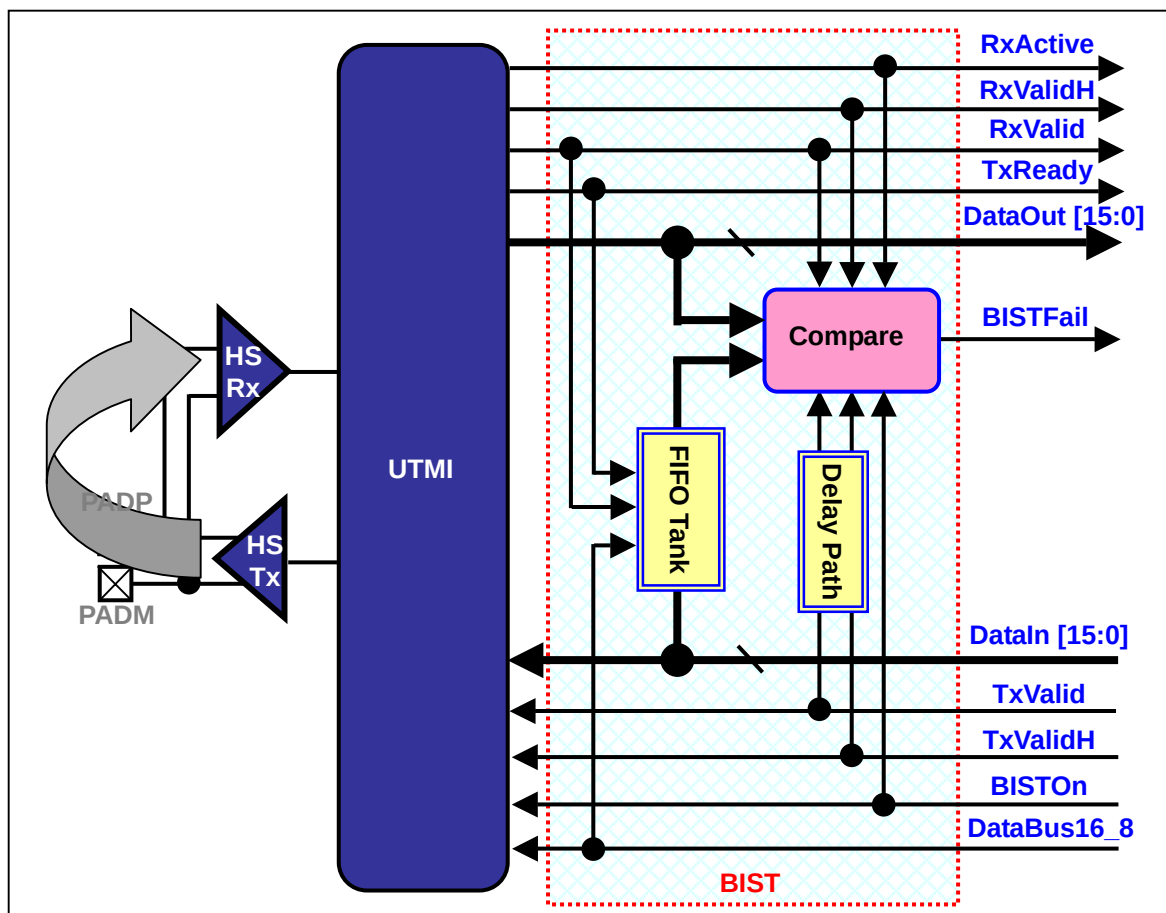


Fig-5 Function diagram of UTM built-in-self-test

Fig-5 shows the BIST function diagram of *tpd973g_usb2* UTM. Note that BIST is only suitable for high-speed **OpMode0** test, that is, **XcvrSelect**=0, **TermSelect**=0, and **OpMode[1:0]**=0. For full-speed mode, the output of **BISTFail** is not meaningful. The BIST circuit can work well for both 8-bit and 16-bit bus interface.

When BIST is enabled with **BISTOn**=1, the transmitter to receiver self-loop-back path of digital UTMI and analog PHY is created. So, all the transmitted (**DataIn [15:0]**) test pattern will be compared to the loop-back output (**DataOut [15:0]**) with proper delay. If the received data does not comply with the transmitted data, **BISTFail** flags “high”, and failure is detected on the chip.

3.2 Reduce Test Pin in SoC application

In SoC integration of *tpd973g_usb2* UTM, the fewer multiplexed UTMI test pin is preferred. Users can insert the gate in dash box of Fig. 6 to ensure the high-speed BIST mode is set up properly. In this way, only 21 pins (**DataIn [15:0]**, **TxReady**, **TxValid**, **TxValidH**, **BISTOn**, **BISTFail**) are required for BIST test

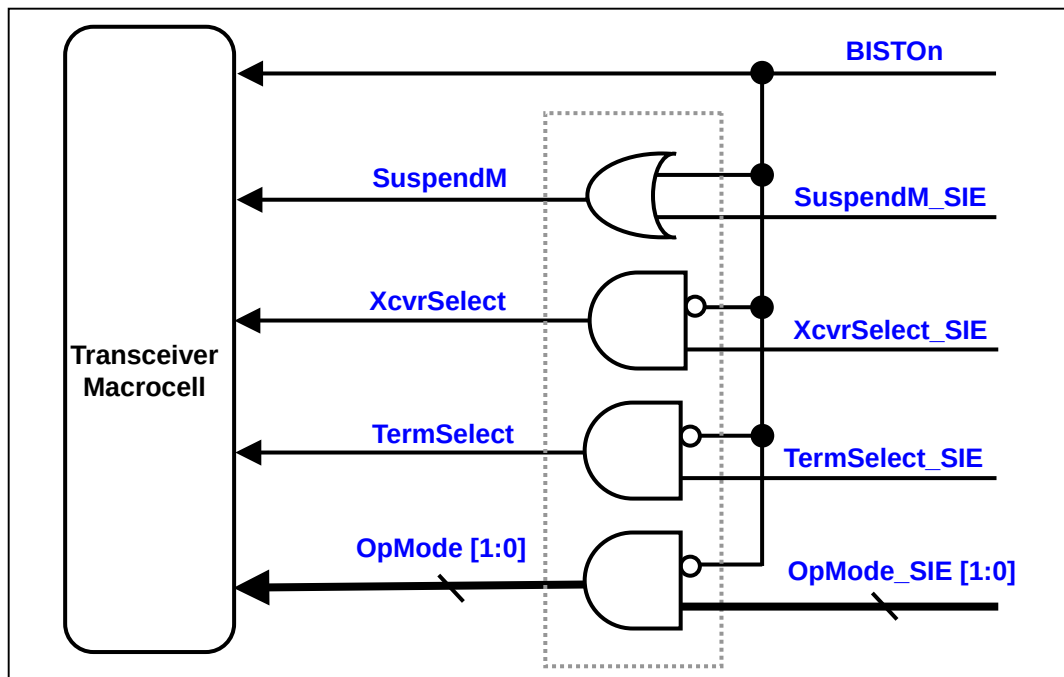


Fig-6 Reduce BIST test pin counts to UTM interface



3.3 BIST Test Pattern

TSMC doesn't provide any pattern testing for BIST. The BIST function just builds a hardware pipeline for users. Basically the pattern's definition and generation are all up to users. If users are not familiar with pattern definition, they can refer to the Universal Serial Bus Specification. Users can edit the pattern as in the following table list(Table-1) to run simple functional tests and judge whether the high speed mode passed or failed.

Table-1 Test Pattern Example

Item No	NRZI Symbols (Fields)	NRZ Bit Strings	Number of NRZ Bits
1	{ KJ *15 }, KK (SYNC)	{ 00000000 * 3 }, 00000001	32
2	KKJKJKKK (DATA0 PID)	11000011	8
3	JKJKJKJK * 9	00000000 * 9	72
4	JJKKJJKK * 8	01010101 * 8	64
5	JJJJJKKK * 8	01110111 * 8	64
6	JJJJJJJKKKKKKK * 8	0, { 111111S * 15 }, 111111	97
7	JJJJJJK * 8	S, 111111S, { 0111111S * 7 }	55
8	{ JKKKKKKK * 10 }, JK	00111111, { S0111111 * 9 }, S0	72
9	JJJKKKJJKKKKJKKK (CRC16)	0110110101110011	16
10	JJJJJJJ (EOP)	01111111	8

* Extracted from Universal Serial Bus Specification rev._2.0 Apr.27, 2000 7.1.20 sec.



4 VERILOG USAGE

Because the *tpd973g_usb2* has complex circuit and logic state, it is hard to use Hspice or any gate level simulator to do verification. TSMC strongly recommend users to use Verilog to verify integration result for PHY and relater SIE and application circuit. The *tpd973g_usb2* provides two encrypted Verilog models for diffenent tools: one is Verilog XL/NCL™ for Cadence™, the other is ModelSim™ for Mentor Graphics™. Both encrypted Verilog models include behavioral RTL code(pre-simulation) as well as timing synthesized GTL code(post-simulation).

4.1 General Questions

4.1.1 Abnormal condition

In full speed mode, and the device is set in RX mode **RxActive**, **RxValid** and **RxValidH** may not be active to make receiver function fail. The failure is a false error. The reason is caused by Verilog behavior model detecting the unknown internal RXDFS node at SE0 mode. To fix this false error, users just need to shift the valid RX data with approximate 1nS earlier and later at **PADP/PADM** pins. This would not occur in real silicon functionality.

4.1.2 ModelSim™ Version problem

Users may face version issue if they use ModelSim™ as Verilog simulator. Depending on ModelSim™ version they choose, users may need to regenerate Verilog libraries before running simulation. Please read the installation README file of ModelSim™ to see if regeneration for *tpd973g_usb2* is necessary. There are two solutions to regenerate Verilog library. Users can regenerate *tpd973g_usb2* libraries by using the Refresh command from the Library tab context menu, or by using the `–refresh` argument to `vlog`.

From the command line, users would use `vlog` with the `–refresh` option to update Verilog design units. By default, the work library is updated; use `–work <library>` to update a different library. For example, the USB2 library named *tpd973g_usb2* that contains Verilog design units.

```
vlog –work tpd973g_usb2 –refresh
```



An important feature of –refresh is that it rebuilds the library image without using source code. This means that models delivered as compiled libraries without source code can be rebuilt for a specific release of ModelSim™(4.6 and later only). In general, this works for moving forwards or backwards on a release. Moving backwards on a release may not work. For any problem in usage about ModelSim™, please contact Mentor Graphics™ AE.



5 OPERATION AND TEST

5.1 Suspend mode

When users enable the suspend mode, they should let **SUSPENDM** low and active. In this mode, the system clock (**CLK**) is stopped and kept in high state, until users want to depart from suspend mode by setting **SUSPENDM** to high state. At that moment the system clock (**CLK**) will be still kept to high state, and will stay in high state for about 3mS without any clock generating. (USB2.0 spec. specifies that period of time should be less than 5.6mS) Users who want to input any pattern from SIE, should wait 3mS after **SUSPENDM** is set back to high.

For users' convenience, the 3mS delay time in physical level will be re-set to 30nS in Verilog model.

For the user who wants to stop the clock/oscillator in suspend state must note this description. For the simple and easy usage TSMC sets the reset automation before suspend. It means the USB2.0 doesn't need to set the reset pin but it will reset all the circuit inside when user directly pulls the **SUSPENDM** pin to low. If user for decreasing the power consumption needs to stop the clock(**FREF**), user must keep the input **FREF** clock toggling a while after suspend enabling. The safe and the least guideline for **FREF** keeps alive must be longer than 1uS for resetting usage after **SUSPENDM** goes low. If **FREF** clock is stoped early, it has the risk that USB2.0 can't get into suspend mode and some unknown states in some pins will continually exist even off the suspend. When USB2.0 is on from suspend mode, of cause the clock(**FREF**) must be normal when the USB2.0 gets away from suspend mode. TSMC reminds the user the clock should be lower than 500ppm jitter when the **SUSPENDM** goes back high state. If user fits the requirement, user can do the normal function after 3mS.

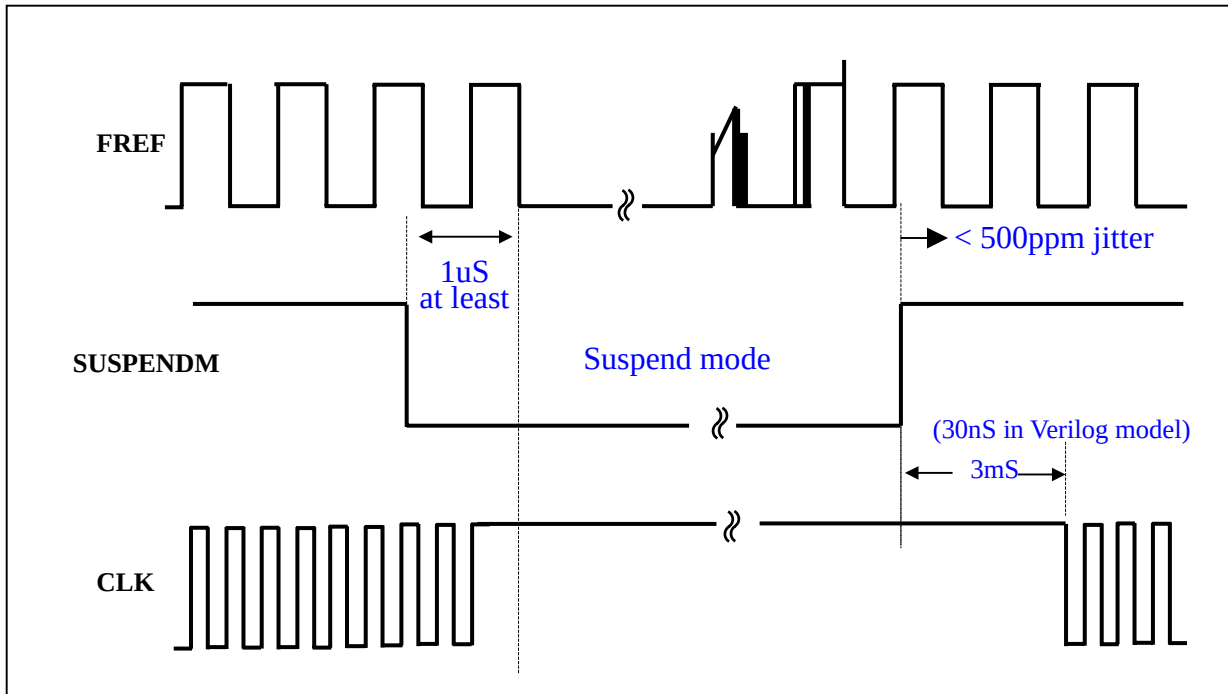


Fig-7 Relation between suspend signal and clock timing



5.2 Test

During design stages, users should finish the verilog simulation with TSMC verilog model and the in-house compliance test of TSMC USB2.0 UTMI evaluation board controlled by customer's controller before tape-out. And also customers are encouraged to add on-chip test features to control/monitor the UTMI PHY macro directly outside the chip to double confirmed the UTMI PHY is functional for all compliance test items for the design verification afterwards in real silicon. The detailed compliance test information can be found in the web site of USB organization, www.usb.org.

During production test, users can measure the current consumption of each power for Iddq test and adopt the built-in BIST function as shown in Section 3.3 for high speed AC testing to shorten the test time. However, to characterize all the items in compliance test requires the on-chip test features to control/monitor the UTMI PHY macro directly outside the chip.



6 ESD CONSIDERATION

6.1 Tie-Off UTM pins

In some applications, there are some pins not necessary and should be tie-off. For example, DataBus16_8. Please DO NOT tie-off these pins directly to power or ground, it may cause weak ESD protection. Please connect these pins with the “tie-high” and “tie-low” cells provided in your standard core cell library. ESD performance will be improved with this arrangement.

6.2 Power-Cut Cells

There are built-in power-cut cells in *tpd973g_usb2* UTM. Users need no extra effort to attach power-cut cells from analog I/O library. Note that the redundant power cut cell would degrade the cross-power-domain ESD performance and should be avoided.



7 LAYOUT CONSIDERATION

7.1 Placement & Routing

It is recommended to place this cell near the corner, far away from any other fast switching digital I/O pads or any other I/O pads with high driving capability. Any core circuit's signal/power/ground trace can not cross over USB2.0 transceiver macrocell. Any noisy or fast switching circuit should be away from this macrocell.

It is recommended to leave a void region between this macrocell between other core circuits to reduce unwanted noise coupling effect. Please keep 30um between USB2.0 transceiver marco and core circuit's signal/power/ground, and 100um between core circuits' diffusion and wills.

7.2 Pad Sequence

The USB2.0 macrocell already include bonding pads, so users don't need to add them by themselves. If they use the phantom view to layout, like Apollo™ FRAM view(apf) or Silicon Ensemble™ LEF view(sef), then they will see macrocell block(blue twill in Fig-8) but won't see the pad area(pink rectangle in Fig-8). However, for users' convenience, TSMC provides the coordinates of the bonding pads as follows:

Macrocell original point for **PUSB20TM** : (0 , 0)

Pad original point: (x , y) at every pad's left-bottom side.

	PrBoundary: drawing[235:252]	PAD: drawing[19:252]
1. VDDA :	(35.000 , -75.500)	(38.500 , -73.500)
2. RPU :	(107.000 , -75.500)	(110.500 , -73.500)
3. PADP :	(265.890 , -75.500)	(269.390 , -73.500)
4. PADM :	(335.560 , -75.500)	(339.060 , -73.500)
5. VSSA :	(480.075 , -75.500)	(483.575 , -73.500)
6. VDDD :	(822.805 , -75.500)	(826.305 , -73.500)
7. VDDL :	(894.805 , -75.500)	(898.305 , -73.500)

8. VSDL :	(966.805 , -75.500)	(970.305 , -73.500)
9. VDDA :	(1038.805 , -75.500)	(1042.305 , -73.500)
A. VSSA :	(1110.805 , -75.500)	(1114.305 , -73.500)
B. VRES :	(1182.805 , -75.500)	(1186.305 , -73.500)

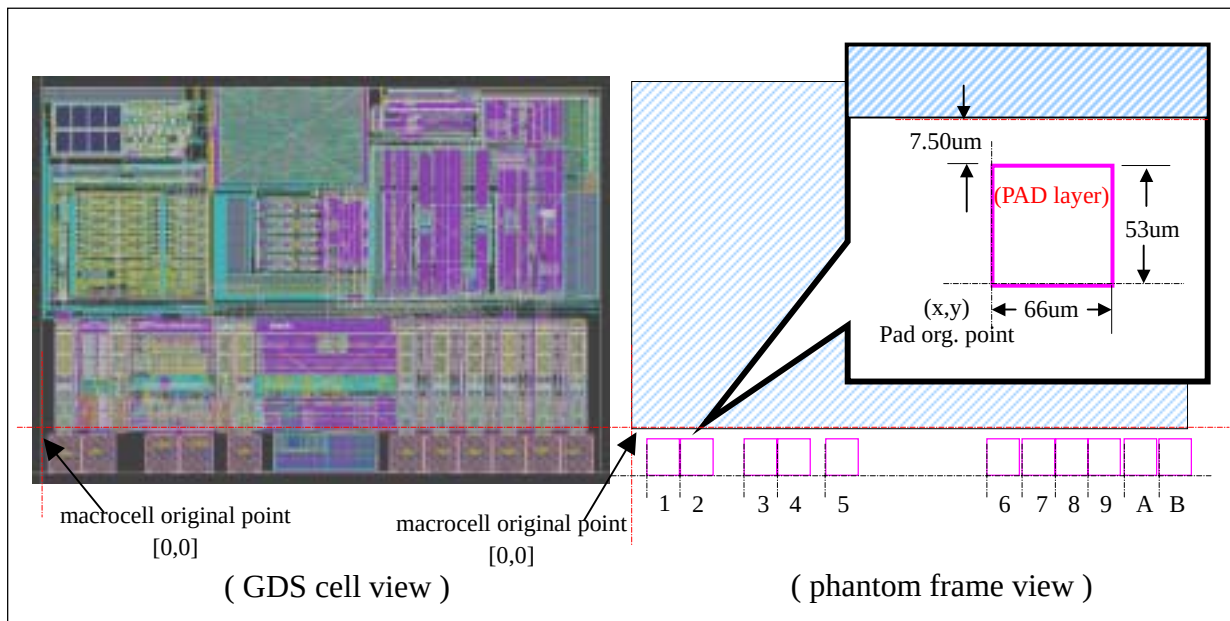


Fig-8 Contrast with Phantom view and gds